

TrES-2b

R-2097

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120425 · created 2026-08-10T22:59:14+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456042.83294 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1491 +/- 0.008
Transit depth (Rp/Rs)^2	2.22 +/- 0.24 [%]
Orbital Inclination (inc)	83.76 +/- 0.26
Airmass coefficient 1 (a1)	1.012 +/- 0.0094
Airmass coefficient 2 (a2)	-0.0081 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#3 - [132, 273]
Optimal Aperture	4.28
Optimal Annulus	8.53
Transit Duration (day)	0.0764 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1491 ± 0.008 vs 0.1254 (deviation 2.86σ)
Duration fit vs published	0.0764 d vs 0.0746 d (deviation 0.42σ)
Mid-time O–C	-0.4 min
Depth SNR	18.0
Residual scatter	0.92 %

Light curve

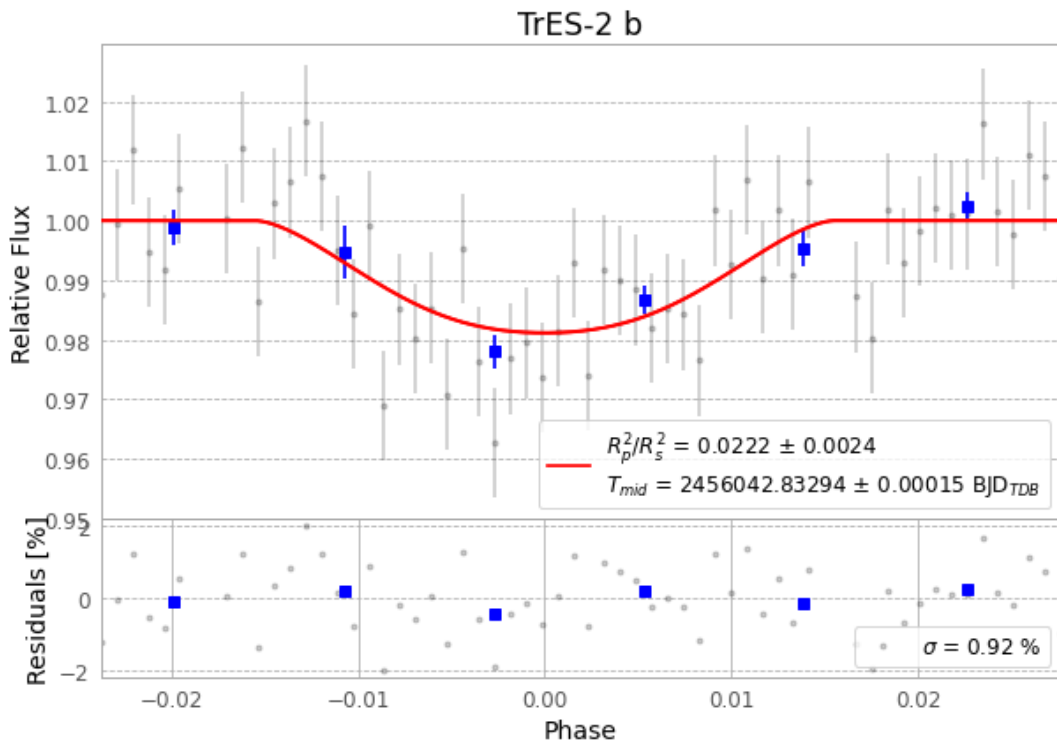


Figure 1 — FinalLightCurve_TrES-2 b_2012-04-24.png (SHA-256 2d4b13edb87fc38d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 142], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 595deb35787a400f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), TRES-2120425063312.FITS → TRES-2120425094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120425.log (41c3254af10e...), FinalLightCurve_TrES-2 b_2012-04-24.png (2d4b13edb87f...), FinalLightCurve_TrES-2 b_2012-04-24.csv (34bfc5630696...)

Compute resources

Wall time 429.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 22:59Z.